

Review Questions

The answers to the chapter review questions can be found in the Appendix.

1. Which of the following Java operators can be used with `boolean` variables? (Choose all that apply.)

A. `==`

B. `+`

C. `--`

D. `!`

E. `%`

F. `~`

G. Cast with `(boolean)`

2. What data type (or types) will allow the following code snippet to compile? (Choose all that apply.)

```
byte apples = 5;  
short oranges = 10;  
_____ bananas = apples + oranges;
```

A. `int`

B. `long`

C. `boolean`

D. `double`

E. `short`

F. `byte`

3. What change, when applied independently, would allow the following code snippet to compile? (Choose all that apply.)

```
3: long ear = 10;  
4: int hearing = 2 * ear;
```

A. No change; it compiles as is.

B. Cast `ear` on line 4 to `int`.

C. Change the data type of `ear` on line 3 to `short`.

D. Cast `2 * ear` on line 4 to `int`.

E. Change the data type of `hearing` on line 4 to `short`.

F. Change the data type of `hearing` on line 4 to `long`.

4. What is the output of the following code snippet?

```
3: boolean canine = true, wolf = true;
4: int teeth = 20;
5: canine = (teeth != 10) ^ (wolf=false);
6: System.out.println(canine+", "+teeth+", "+wolf);
```

- A. true, 20, true
- B. true, 20, false
- C. false, 10, true
- D. false, 20, false
- E. The code will not compile because of line 5.
- F. None of the above.

5. Which of the following operators are ranked in increasing or the same order of precedence? Assume the + operator is binary addition, not the unary form. (Choose all that apply.)

- A. +, *, %, --
- B. ++, (int), *
- C. =, ==, !
- D. (short), =, !, *
- E. *, /, %, +, ==
- F. !, |, &
- G. ^, +, =, +=

6. What is the output of the following program?

```
1: public class CandyCounter {
2:     static long addCandy(double fruit, float vegetables) {
3:         return (int)fruit+vegetables;
4:     }
5:
6:     public static void main(String[] args) {
7:         System.out.print(addCandy(1.4, 2.4f) + ", ");
8:         System.out.print(addCandy(1.9, (float)4) + ", ");
9:         System.out.print(addCandy((long)(int)(short)2, (float)4)); } }
```

- A. 4, 6, 6.0
- B. 3, 5, 6
- C. 3, 6, 6
- D. 4, 5, 6
- E. The code does not compile because of line 9.
- F. None of the above.

7. What is the output of the following code snippet?

```
int ph = 7, vis = 2;
boolean clear = vis > 1 & (vis < 9 || ph < 2);
boolean safe = (vis > 2) && (ph++ > 1);
boolean tasty = 7 <= --ph;
System.out.println(clear + "-" + safe + "-" + tasty);
```

- A. true-true-true
- B. true-true-false
- C. true-false-true
- D. true-false-false
- E. false-true-true
- F. false-true-false
- G. false-false-true
- H. false-false-false

8. What is the output of the following code snippet?

```
4: int pig = (short)4;
5: pig = pig++;
6: long goat = (int)2;
7: goat -= 1.0;
8: System.out.print(pig + " - " + goat);
```

- A. 4 - 1
- B. 4 - 2
- C. 5 - 1
- D. 5 - 2
- E. The code does not compile due to line 7.
- F. None of the above.

9. What are the unique outputs of the following code snippet? (Choose all that apply.)

```
int a = 2, b = 4, c = 2;
System.out.println(a > 2 ? --c : b++);
System.out.println(b = (a!=c ? a : b++));
System.out.println(a > b ? b < c ? b : 2 : 1);
```

- A. 1
- B. 2
- C. 3
- D. 4
- E. 5
- F. 6
- G. The code does not compile.

10. What are the unique outputs of the following code snippet? (Choose all that apply.)

```
short height = 1, weight = 3;
short zebra = (byte) weight * (byte) height;
double ox = 1 + height * 2 + weight;
long giraffe = 1 + 9 % height + 1;
System.out.println(zebra);
System.out.println(ox);
System.out.println(giraffe);
```

- A. 1
- B. 2
- C. 3
- D. 4
- E. 5
- F. 6
- G. The code does not compile.

11. What is the output of the following code?

```
11: int sample1 = (2 * 4) % 3;
12: int sample2 = 3 * 2 % 3;
13: int sample3 = 5 * (1 % 2);
14: System.out.println(sample1 + ", " + sample2 + ", " + sample3);
```

- A. 0, 0, 5
- B. 1, 2, 10
- C. 2, 1, 5
- D. 2, 0, 5
- E. 3, 1, 10
- F. 3, 2, 6
- G. The code does not compile.

12. The _____ operator increases a value and returns the original value, while the _____ operator decreases a value and returns the new value.

- A. post-increment, post-increment
- B. pre-decrement, post-decrement
- C. post-increment, post-decrement
- D. post-increment, pre-decrement
- E. pre-increment, pre-decrement
- F. pre-increment, post-decrement

13. What is the output of the following code snippet?

```
boolean sunny = true, raining = false, sunday = true;
boolean goingToTheStore = sunny & raining ^ sunday;
boolean goingToTheZoo = sunday && !raining;
boolean stayingHome = !(goingToTheStore && goingToTheZoo);
System.out.println(goingToTheStore + "-" + goingToTheZoo
    + "-" +stayingHome);
```

- A. true-false-false
 - B. false-true-false
 - C. true-true-true
 - D. false-true-true
 - E. false-false-false
 - F. true-true-false
 - G. None of the above
14. Which of the following statements are correct? (Choose all that apply.)
- A. The return value of an assignment operation expression can be void.
 - B. The inequality operator (!=) can be used to compare objects.
 - C. The equality operator (==) can be used to compare a boolean value with a numeric value.
 - D. During runtime, the & and | operators may cause only the left side of the expression to be evaluated.
 - E. The return value of an assignment operation expression is the value of the newly assigned variable.
 - F. In Java, 0 and false may be used interchangeably.
 - G. The logical complement operator (!) cannot be used to flip numeric values.
15. Which operators take three operands or values? (Choose all that apply.)
- A. =
 - B. &&
 - C. *=
 - D. ? :
 - E. &
 - F. ++
 - G. /

16. How many lines of the following code contain compiler errors?

```
int note = 1 * 2 + (long)3;  
short melody = (byte)(double)(note *= 2);  
double song = melody;  
float symphony = (float)((song == 1_000f) ? song * 2L : song);
```

- A. 0
- B. 1
- C. 2
- D. 3
- E. 4

17. Given the following code snippet, what are the values of the variables after it is executed? (Choose all that apply.)

```
int ticketsTaken = 1;  
int ticketsSold = 3;  
ticketsSold += 1 + ticketsTaken++;  
ticketsTaken *= 2;  
ticketsSold += (long)1;
```

- A. ticketsSold is 8.
- B. ticketsTaken is 2.
- C. ticketsSold is 6.
- D. ticketsTaken is 6.
- E. ticketsSold is 7.
- F. ticketsTaken is 4.
- G. The code does not compile.

18. Which of the following can be used to change the order of operation in an expression? (Choose all that apply.)

- A. []
- B. < >
- C. ()
- D. \ /
- E. { }
- F. " "

19. What is the result of executing the following code snippet? (Choose all that apply.)

```
3: int start = 7;  
4: int end = 4;  
5: end += ++start;  
6: start = (byte)(Byte.MAX_VALUE + 1);
```

- A. start is 0.
 - B. start is -128.
 - C. start is 127.
 - D. end is 8.
 - E. end is 11.
 - F. end is 12.
 - G. The code does not compile.
 - H. The code compiles but throws an exception at runtime.
20. Which of the following statements about unary operators are true? (Choose all that apply.)
- A. Unary operators are always executed before any surrounding numeric binary or ternary operators.
 - B. The `~` operator can be used to flip a boolean value.
 - C. The pre-increment operator (`++`) returns the value of the variable before the increment is applied.
 - D. The post-decrement operator (`--`) returns the value of the variable before the decrement is applied.
 - E. The `!` operator cannot be used on numeric values.
 - F. None of the above
21. What is the result of executing the following code snippet?
- ```
int myFavoriteNumber = 8;
int bird = ~myFavoriteNumber;
int plane = -myFavoriteNumber;
var superman = bird == plane ? 5 : 10;
System.out.println(bird + "," + plane + "," + --superman);
```
- A. -7,-8,9
  - B. -7,-8,10
  - C. -8,-8,4
  - D. -8,-8,5
  - E. -9,-8,9
  - F. -9,-8,10
  - G. None of the above